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problem1

To calculate the probability and properties of the random variables X and Y , we first define the region S where the joint PDF is non-zero. The region $|x| + |y| \leq 2$ describes a diamond shape with vertices at $(2, 0)$, $(0, 2)$, $(-2, 0)$, and $(0, -2)$.

The area of this region is $1/2 \times 4 \times 4 = 8$. Since the PDF $f_{XY}(x, y) = 1/8$ for $(x, y) \in S$ and $\int_S (1/8) dA = (1/8) \times 8 = 1$, the PDF is properly normalized.

a) Calculate $P(A)$

The event A is defined by $y \geq x^2$. Since $x^2 \geq 0$, we only consider the portion of the diamond in the upper half-plane ($y \geq 0$). In the upper half-plane, the diamond boundary is $y \leq 2 - |x|$.

The region of interest is the intersection of $y \geq x^2$ and $y \leq 2 - |x|$. To find the integration limits for x , we set $x^2 = 2 - |x|$:

$$x^2 + |x| - 2 = 0 \implies (|x| + 2)(|x| - 1) = 0 \implies |x| = 1 \implies x = \pm 1$$

The area of region A is:

$$\begin{aligned} \text{Area}(A) &= \int_{-1}^1 \int_{x^2}^{2-|x|} dy dx = 2 \int_0^1 (2 - x - x^2) dx \\ \text{Area}(A) &= 2 \left[2x - \frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = 2 \left(2 - \frac{1}{2} - \frac{1}{3} \right) = 2 \left(\frac{12 - 3 - 2}{6} \right) = \frac{7}{3} \end{aligned}$$

The probability is the area of A scaled by the constant PDF value:

$$P(A) = \text{Area}(A) \cdot \frac{1}{8} = \frac{7}{3} \cdot \frac{1}{8} = \frac{7}{24}$$

b) Find $f_Y(y|A)$

The conditional PDF is $f_Y(y|A) = \frac{f_Y(y,A)}{P(A)}$, where $f_Y(y,A) = \int f_{XY}(x,y) \mathbb{I}((x,y) \in A) dx$. For a fixed y , x is constrained by $|x| \leq \sqrt{y}$ (from $y \geq x^2$) and $|x| \leq 2-y$ (from the diamond).

1. **If $0 \leq y \leq 1$:** $\sqrt{y} \leq 2-y$, so $|x| \leq \sqrt{y}$.

$$f_Y(y,A) = \int_{-\sqrt{y}}^{\sqrt{y}} \frac{1}{8} dx = \frac{2\sqrt{y}}{8} = \frac{\sqrt{y}}{4}$$

2. **If $1 < y \leq 2$:** $2-y < \sqrt{y}$, so $|x| \leq 2-y$.

$$f_Y(y,A) = \int_{-(2-y)}^{2-y} \frac{1}{8} dx = \frac{2(2-y)}{8} = \frac{2-y}{4}$$

Dividing by $P(A) = 7/24$:

$$f_Y(y|A) = \begin{cases} \frac{6}{7}\sqrt{y} & 0 \leq y \leq 1 \\ \frac{6}{7}(2-y) & 1 < y \leq 2 \end{cases}$$

c) Compute $\text{var}(X|A)$

Due to the symmetry of region A and the PDF across the y -axis, the conditional mean $E[X|A] = 0$. Thus, $\text{var}(X|A) = E[X^2|A]$.

$$E[X^2, A] = \iint_A x^2 \frac{1}{8} dA = \frac{2}{8} \int_0^1 x^2(2-x-x^2) dx$$

$$E[X^2, A] = \frac{1}{4} \left[\frac{2x^3}{3} - \frac{x^4}{4} - \frac{x^5}{5} \right]_0^1 = \frac{1}{4} \left(\frac{2}{3} - \frac{1}{4} - \frac{1}{5} \right) = \frac{1}{4} \left(\frac{40-15-12}{60} \right) = \frac{13}{240}$$

Normalizing by $P(A)$:

$$\text{var}(X|A) = \frac{E[X^2, A]}{P(A)} = \frac{13/240}{7/24} = \frac{13}{240} \cdot \frac{24}{7} = \frac{13}{70}$$

problem2

To solve for the properties of the jointly normal random variables X and Y , we first summarize the given parameters in vector and matrix form.

The mean vector is:

$$M_X = \begin{bmatrix} E[X] \\ E[Y] \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

The covariance matrix C_X is composed of the variances on the diagonal and the covariance on the off-diagonals:

$$C_X = \begin{bmatrix} \text{var}(X) & \text{cov}(X,Y) \\ \text{cov}(X,Y) & \text{var}(Y) \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

a) Explicit Expression for $f_{XY}(x, y)$

The probability density function (PDF) for a bivariate normal distribution is given by:

$$f_{XY}(x, y) = \frac{1}{2\pi\sqrt{\det(C_X)}} \exp\left(-\frac{1}{2}(X - M_X)^T C_X^{-1}(X - M_X)\right)$$

1. **Calculate the determinant:** $\det(C_X) = (2)(2) - (-1)(-1) = 3$.
2. **Find the inverse matrix:** $C_X^{-1} = \frac{1}{3} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$.
3. **Expand the exponent:**

$$\begin{aligned} (X - M_X)^T C_X^{-1}(X - M_X) &= \frac{1}{3} \begin{bmatrix} x-1 & y-1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x-1 \\ y-1 \end{bmatrix} \\ &= \frac{1}{3} [2(x-1)^2 + 2(x-1)(y-1) + 2(y-1)^2] \end{aligned}$$

Substituting these into the PDF formula:

$$f_{XY}(x, y) = \frac{1}{2\pi\sqrt{3}} \exp\left(-\frac{1}{3} [(x-1)^2 + (x-1)(y-1) + (y-1)^2]\right)$$

b) Covariance Ellipse Sketch

The covariance ellipse is centered at $(1, 1)$. Its orientation and shape are determined by the eigenvalues and eigenvectors of C_X .

1. **Eigenvalues:** Solving $\det(C_X - \lambda I) = 0 \implies (2 - \lambda)^2 - 1 = 0$ gives $\lambda_1 = 3$ and $\lambda_2 = 1$.
2. **Eigenvectors:**

- For $\lambda_1 = 3$, the eigenvector is $E_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ (slope = -1).
- For $\lambda_2 = 1$, the eigenvector is $E_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ (slope = 1).

Sketch Description: * **Center:** $(1, 1)$. * **Major Axis:** Along the line $y - 1 = -(x - 1)$ (or $y = -x + 2$). The ellipse is elongated in this direction because $\lambda_1 = 3$ is the larger eigenvalue. * **Minor Axis:** Along the line $y - 1 = (x - 1)$ (or $y = x$). * **Orientation:** The negative covariance causes the ellipse to tilt downward from top-left to bottom-right.

c) Orthogonal Matrix U for Decorrelation

To make the random vector $W = U[X, Y]^T$ uncorrelated, the rows of the orthogonal matrix U must be the normalized eigenvectors of C_X .

$$U = \begin{bmatrix} E_2^T \\ E_1^T \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

Using this transformation, the new covariance matrix C_W will be diagonal:

$$C_W = U C_X U^T = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$$

d) Compute $E[W]$

Using the linearity of expectation:

$$\begin{aligned} E[W] &= U E \begin{bmatrix} X \\ Y \end{bmatrix} = U \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ E[W] &= \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{2}{\sqrt{2}} \\ 0 \end{bmatrix} = \begin{bmatrix} \sqrt{2} \\ 0 \end{bmatrix} \end{aligned}$$

problem3

To solve the problem for the complex random 3-vector Z , we rely on the fundamental properties of complex covariance matrices and linear transformations.

a) Find the values of a, b , and c

A covariance matrix C_Z of a complex random vector must be **Hermitian**. By definition, a Hermitian matrix satisfies $C_Z = C_Z^\dagger$, where $C_Z^\dagger$ is the conjugate transpose (the transpose of the matrix with each element replaced by its complex conjugate).

The given covariance matrix is:

$$C_Z = \begin{bmatrix} 4 & i & a \\ b & 3 & 0 \\ 1 & c & 3 \end{bmatrix}$$

Applying the Hermitian property $C_{jk} = C_{kj}^*$: 1. **For** b : C_{21} must be the conjugate of C_{12} .

$$b = C_{21} = C_{12}^* = (i)^* = -i$$

2. **For** a : C_{13} must be the conjugate of C_{31} .

$$a = C_{13} = C_{31}^* = (1)^* = 1$$

3. **For** c : C_{32} must be the conjugate of C_{23} .

$$c = C_{32} = C_{23}^* = (0)^* = 0$$

Thus, the values are $a = 1$, $b = -i$, and $c = 0$. The resulting covariance matrix is:

$$C_Z = \begin{bmatrix} 4 & i & 1 \\ -i & 3 & 0 \\ 1 & 0 & 3 \end{bmatrix}$$

b) Calculate $\text{var}(X)$

For a complex scalar random variable X defined as a linear combination of the components of Z , $X = \mathbf{a}^T Z$, the variance is given by the quadratic form:

$$\text{var}(X) = \mathbf{a}^T C_Z \mathbf{a}^*$$

where $\mathbf{a}^*$ is the complex conjugate of the coefficients.

Given $X = 2z_1 - 3z_2 + (1 + 2i)z_3$, we define the coefficient vector:

$$\mathbf{a}^T = [2 \quad -3 \quad 1 + 2i]$$

We calculate the variance in two steps:

Step 1: Calculate $\mathbf{a}^T C_Z$

$$\begin{aligned} \mathbf{a}^T C_Z &= [2 \quad -3 \quad 1 + 2i] \begin{bmatrix} 4 & i & 1 \\ -i & 3 & 0 \\ 1 & 0 & 3 \end{bmatrix} \\ &= [2(4) + (-3)(-i) + (1 + 2i)(1) \quad 2(i) + (-3)(3) + (1 + 2i)(0) \quad 2(1) + (-3)(0) + (1 + 2i)(3)] \\ &= [8 + 3i + 1 + 2i \quad 2i - 9 \quad 2 + 3 + 6i] \end{aligned}$$

$$= \begin{bmatrix} 9 + 5i & -9 + 2i & 5 + 6i \end{bmatrix}$$

Step 2: Multiply by a*

$$\text{var}(X) = \begin{bmatrix} 9 + 5i & -9 + 2i & 5 + 6i \end{bmatrix} \begin{bmatrix} 2 \\ -3 \\ 1 - 2i \end{bmatrix}$$

$$\begin{aligned} \text{var}(X) &= (9 + 5i)(2) + (-9 + 2i)(-3) + (5 + 6i)(1 - 2i) \\ &= (18 + 10i) + (27 - 6i) + (5 - 10i + 6i - 12i^2) \\ &= 18 + 10i + 27 - 6i + 5 - 4i + 12 \\ &= (18 + 27 + 5 + 12) + (10 - 6 - 4)i \\ \text{var}(X) &= 62 \end{aligned}$$

The variance of X is **62**.

problem4

To calculate the bias and variance of the estimator $\hat{\theta} = \frac{2}{n} \sum_{k=1}^n x_k$, we first identify the statistical properties of the individual random variables x_k based on the uniform distribution $U[0, \theta]$.

1. Properties of the Uniform Distribution

For a random variable $x_k \sim U[0, \theta]$: * **Mean** ($E[x_k]$): The expected value is the midpoint of the interval:

$$E[x_k] = \frac{0 + \theta}{2} = \frac{\theta}{2}$$

* **Variance** ($\text{var}(x_k)$): The variance is given by $\frac{(b-a)^2}{12}$:

$$\text{var}(x_k) = \frac{(\theta - 0)^2}{12} = \frac{\theta^2}{12}$$

2. Calculation of Bias $\beta(\hat{\theta})$

The bias of an estimator is defined as $\beta(\hat{\theta}) = E[\hat{\theta}] - \theta$. First, we find the expected value of the estimator:

$$E[\hat{\theta}] = E \left[\frac{2}{n} \sum_{k=1}^n x_k \right]$$

By the linearity of expectation:

$$\begin{aligned} E[\hat{\theta}] &= \frac{2}{n} \sum_{k=1}^n E[x_k] = \frac{2}{n} \left(n \cdot \frac{\theta}{2} \right) \\ E[\hat{\theta}] &= \theta \end{aligned}$$

Since $E[\hat{\theta}] = \theta$, the bias is:

$$\beta(\hat{\theta}) = \theta - \theta = 0$$

The estimator $\hat{\theta}$ is an **unbiased** estimator of θ .

3. Calculation of Variance $var(\hat{\theta})$

The variance of the estimator is calculated using the property that for independent random variables, the variance of their sum is the sum of their variances:

$$var(\hat{\theta}) = var\left(\frac{2}{n} \sum_{k=1}^n x_k\right)$$

Applying the property $var(aX) = a^2 var(X)$:

$$var(\hat{\theta}) = \left(\frac{2}{n}\right)^2 var\left(\sum_{k=1}^n x_k\right)$$

Since $x_1, \dots, x_n$ are iid:

$$\begin{aligned} var(\hat{\theta}) &= \frac{4}{n^2} \sum_{k=1}^n var(x_k) \\ var(\hat{\theta}) &= \frac{4}{n^2} \left(n \cdot \frac{\theta^2}{12}\right) \end{aligned}$$

Simplifying the expression:

$$var(\hat{\theta}) = \frac{4n\theta^2}{12n^2} = \frac{\theta^2}{3n}$$

Summary of Results

- **Bias:** $\beta(\hat{\theta}) = 0$
- **Variance:** $var(\hat{\theta}) = \frac{\theta^2}{3n}$

problem 5

To find the bias of the new estimator $\bar{\theta}_1$, we use the definition of bias and the linearity of expectation.

1. Definition of Bias

The bias of an estimator $\bar{\theta}$ for a parameter θ is defined as:

$$\text{Bias}(\bar{\theta}) = E[\bar{\theta}] - \theta$$

Given that the bias of $\bar{\theta}$ is β , we can write:

$$E[\bar{\theta}] - \theta = \beta \implies E[\bar{\theta}] = \theta + \beta$$

2. Calculate the Expected Value of $\bar{\theta}_1$

The new estimator is defined as $\bar{\theta}_1 = 2\bar{\theta}$. Using the linearity property of expectation ($E[aX] = aE[X]$), we find:

$$E[\bar{\theta}_1] = E[2\bar{\theta}] = 2E[\bar{\theta}]$$

Substituting the expression for $E[\bar{\theta}]$ from Step 1:

$$E[\bar{\theta}_1] = 2(\theta + \beta) = 2\theta + 2\beta$$

3. Calculate the Bias of $\bar{\theta}_1$

Now, we apply the definition of bias to the new estimator $\bar{\theta}_1$:

$$\text{Bias}(\bar{\theta}_1) = E[\bar{\theta}_1] - \theta$$

Substitute the result from Step 2:

$$\text{Bias}(\bar{\theta}_1) = (2\theta + 2\beta) - \theta$$

$$\text{Bias}(\bar{\theta}_1) = \theta + 2\beta$$

Summary

The bias of the new estimator $\bar{\theta}_1$ is $\theta + 2\beta$.